

Pre-Development Analysis & Implementation Plan

A complete assessment of the project specification and the existing `ovcu` folder, followed by the target architecture, database and API design, roadmap, risks, and the decisions awaiting approval.

CRITICAL FINDING

The `ovcu` folder contains no application code. It is a Claude Design canvas — one 1,939-line HTML mockup plus vendored runtime and image assets. There is no React project, no .NET solution and no database. The prototype is a valuable, validated UI specification, but every part of the target stack is greenfield.

PROJECT

Ovcuprim.az

PRODUCT TYPE

Specialised classifieds marketplace

PRIMARY MARKET

Azerbaijan

TARGET STACK

React · Vite · TypeScript · ASP.NET Core · EF Core

SOURCES ANALYSED

Ovcuprim.az_Frontend_Plan.docx · D:\ovcu (67 files)

DATE

28 August 2026

STATUS

Analysis complete — awaiting approval

IMPLEMENTATION

Not started. No code written.

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HOW THIS DOCUMENT WAS PRODUCED

Every statement below is derived from two sources only: the full text of `Ovcuprim.az_Frontend_Plan.docx` (42,186 bytes, 22 sections, text-only), and a file-by-file inspection of `D:\ovcu` (67 files). Nothing about the folder's contents was assumed. Where the document is silent — backend, database, API, authentication, hosting — the recommendation is explicitly labelled as mine and appears in the approval checklist.

1 Project documentation analysis

1.1 Business purpose

A **specialised classifieds marketplace** for hunting, fishing, camping and outdoor equipment, built for the Azerbaijani market. Explicitly not a general classifieds site: §21 states the platform should feel like the specialised marketplace of Azerbaijan's outdoor community.

Brand thesis (§1, §21): *"As simple as Tap.az, as characterful as an outdoor brand, as premium as a modern marketplace."*

1.2 Target audience

Azerbaijani hunters, anglers, campers and outdoor enthusiasts — both **individual sellers** and **shops/dealers**. §15 states that the mobile experience must be prioritised above desktop.

1.3 Product concept

A C2C + B2C listings platform: free listing placement, moderation before publication, paid promotion (VIP/TOP), shop storefronts, region-based discovery, and SEO content for organic acquisition.

1.4 Functional requirements extracted from the document

#	Area	Requirement
R1	Categories	6 top-level categories with fixed subcategory lists (§5)
R2	Restricted goods	Separate rules, compliance/age control and moderation for legally restricted products (§5)
R3	Homepage	10 defined sections: header, hero, categories, featured, latest, by-category, regions/map, outdoor guide, CTA, footer (§6)
R4	Header	Logo, Categories, Listings, Shops, Blog , search, login/register, "+ New listing" (§7)
R5	Hero	Full-width Azerbaijani nature imagery, large search, quick filters by category/region/price, tagline "Təbiətə çıx. Lazım olanı tap." (§8)
R6	Listing card	Photo, favourite, price, title, category , region, date, limited badges (New/Shop/TOP); 4/3/2 per row on desktop/tablet/mobile (§9)
R7	Listing page	Breadcrumb → category title → result count → filter panel → sort → grid (§10)
R8	Filters	Common: price, region, new/used, seller type, delivery , date , brand . Plus dynamic per-category attributes — rod: length/test/material/brand; tent: capacity/season/weight; clothing: size/gender/season/material (§10)
R9	Search	Autocomplete, results grouped by category, brand recognition , spelling-variant tolerance , recent searches, works together with filters (§11)
R10	Detail page	Gallery, title, price, condition, region, contact + message buttons, description, specs, seller profile, similar listings, favourite, report (§12)
R11	Sellers	Two types: individual and shop. Shop profile: logo, name, description, address, phone, listings, rating, future verified status (§13)

#	Area	Requirement
R12	Create listing	8-step UX: category → photos → title/description → price/condition → category attributes → region/contact → preview → publish (§14)
R13	Mobile	Bottom nav (Home, Search, +Listing, Favourites, Profile), 2-column grid, always-reachable search and filter (§15)
R14	Map	Azerbaijan map showing listing density by region; region filters — Bakı, Sumqayıt, Gəncə, Quba, Qəbələ, Şəki, Lənkəran and districts (§16)
R15	Content / SEO	Outdoor guide articles for organic traffic and retention (§17)
R16	MVP scope	14 items including registration/login, profile, favourites, seller profile, basic moderation UI , SEO basics (§19)
R17	Phase 2	Shops, verified seller, VIP/TOP, payments, internal messaging, push/email, map, blog, listing statistics, ad system (§20)

1.5 Design system defined by the document (§3–§4)

Token	Value	Documented use
Forest Green	#1F3A32	Primary brand, header, main CTA
Deep Green	#142923	Header/footer, dark surfaces
Olive	#6F7F45	Secondary accent, category elements
Sand	#D9C9A3	Decorative outdoor accent
Warm White	#F7F6F1	Page background
White	#FFFFFF	Cards and content surfaces
Charcoal	#202522	Text
Orange	#D97732	New listing, featured CTA — used sparingly
Typography	Manrope 700 for headings; Inter 400/500 for body	
Radii	Card 12px · button 8px · input 10px · borders #E4E7E2 , minimal shadows	

Explicit constraint from §3: **do not build an all-green or camouflage UI** — it would read as dated and militaristic.

1.6 Contradictions, gaps and risks found in the documentation

These are stated rather than silently assumed away.

ID	Type	Finding
D1	Gap	The document is titled “Frontend and UI/UX Plan”. It defines no backend, no database, no API, no authentication and no hosting . Steps 4–7 of the brief therefore have no documented baseline — every backend decision in this plan is a recommendation, not a documented requirement.
D2	Contradiction	§7 places Blog in the main header and §6 places an outdoor guide on the homepage, but §20 defers the outdoor blog to Phase 2.

ID	Type	Finding
D3	Contradiction	§13 describes shop profiles as a core seller model and §6 puts Shops in the header, but §20 defers shop profiles and verified sellers to Phase 2. §19 lists only “seller profile” in the MVP.
D4	Contradiction	§12 specifies a message button on the detail page; §20 defers internal messaging to Phase 2. A button that opens nothing is worse than no button.
D5	Contradiction	§6 lists featured/VIP listings and monetisation as homepage section 04, but §20 defers VIP/TOP and payment integration to Phase 2.
D6	Incomplete	§5 requires “separate rules, compliance/age control and moderation mechanisms” for legally restricted products but defines none. Hunting firearms, their parts and certain knives are legally regulated in Azerbaijan. This is the highest-risk requirement in the document and it is one sentence long.
D7	Incomplete	§11 requires “spelling variant support”. Azerbaijani has ə/e, ı/i, ş/s, ç/c, ğ/g, ö/o, ü/u, plus Russian-keyboard and Latin-transliteration input. The required tolerance level materially changes the search implementation.
D8	Incomplete	§11 requires “brand recognition” but no brand taxonomy exists. Free text, moderated dictionary, or a per-category entity?
D9	Incomplete	§16 requires a map of Azerbaijan by region. No geometry source is named, and it is unstated whether listings carry coordinates or only a region.
D10	Incomplete	§10 requires per-category dynamic filters but gives three examples. The attribute schema for 40+ subcategories does not exist and must be authored as content.
D11	Risky	§17 and §19 require SEO in the MVP, while §18 prescribes a client-side React SPA structure. A classifieds SPA without server rendering will not rank. This is a genuine architectural conflict inside the document.
D12	Unclear	No language requirement is stated, yet the mockup contains a “Русский язык” switch. Azerbaijani-only versus AZ+RU changes schema, routing and content workload.
D13	Unclear	§19 requires a “basic moderation UI”, but no admin panel, role model or moderator workflow is described anywhere.
D14	Unclear	No listing lifecycle rules appear in the document. The <i>mockup</i> ’s help text defines them (30 days, 10 images × 5 MB, monthly free quota) — that content is currently the only source of truth for these business rules.
D15	Duplication	§5 lists Optika and Bıçaqlar as subcategories of Ovçuluq, and lists “Fənər” under both Kamp and Çanta/aksesuar and “Multitool” under both Ovçuluq and Çanta. The tree needs de-duplication before data entry.
D16	Legal	§2 and §22 name Tap.az, Kamping.az, Campspace and KampçıyızBiz as benchmarks. The document correctly forbids copying visual identity — but see finding M9, where the mockup’s legal text appears adapted from another platform.

2 ovcu folder analysis

2.1 Actual contents

```
D:\ovcu\
└─ Ovçuluq satış saytı tasarımı\
    └─ Ovcuprim.dc.html      154 KB, 1,939 lines  ← the entire prototype
    └─ ListingCard.dc.html   3 KB, 37 lines    ← the one extracted component
    └─ support.js            69 KB (generated canvas runtime – "do not edit")
    └─ image-slot.js         65 KB (vendored starter component)
    └─ .image-slots.state.json 350 KB (one embedded image: key "cat-ov")
    └─ .thumbnail            14 KB
    └─ _ds\modernist-...\    design system: styles.css, readme.md, manifest
    └─ scraps\               document.xml + plan.txt (the docx, extracted)
    └─ uploads\              1 docx + 40 PNGs (25 "draw-*", 15 "pasted-*")

67 files total, 0 lines of application code
← a Claude Design canvas project
```

Searched for and confirmed absent: `package.json`, `*.csproj`, `*.sln`, `*.ts`, `*.tsx`, `*.jsx`, `*.cs`. Zero matches. The docx in `uploads/` is byte-identical to the supplied plan (MD5 `f0b5398d...7464`).

2.2 What the mockup actually is

`Ovcuprim.dc.html` is a single-file reactive component (`class Component extends DCLogic`) using a proprietary template syntax — `<sc-if>`, `<sc-for>`, `<dc-import>`, `{{ binding }}`. All markup is inline-styled. All twelve “pages” live in the same file and are switched by a `state.route` string: **there is no router and no URL changes**.

Route	Line	Coverage in the specification
home	239	§6 — partially
listings	337	§10
detail	454	§12
favorites	569	§19
create	590	§14
ads	781	<i>Not in the document — invented by the mockup</i>
help	861	<i>Not in the document — invented by the mockup</i>
info	891	<i>Not in the document — invented by the mockup</i>
stores	915	§13
store	961	§13
profile	1012	§19
mobile bottom nav	1071	§15

2.3 Mock data inventory

- `CATS` — 8 categories with id, name, initial, fabricated counts and subcategory arrays (67 subcategories in total)
- `REGIONS` — 12 regions with fabricated counts
- `L` — 24 hardcoded listings: id, title, price, category, subcategory, region, age in days, condition, seller type, seller name, TOP flag, spec pairs
- `ATTRS` — per-category attribute label/placeholder pairs for 8 categories plus a fallback
- `HELP` — 6 FAQ groups, 22 question/answer pairs of substantial, well-written Azerbaijani copy
- `INFO` — 12 static pages including a full privacy policy and user agreement
- `ICONS`, `PHOTO_IDS`, `TIMES`, `CONDS`, `SELLERS`, `SORTS`, `WIZ`

2.4 What genuinely works in the prototype

Against the 24 mock rows, entirely client-side: filtering across category, subcategory, region, condition, seller type and min/max price plus free text; three sort modes; favourites; search autocomplete with grouped category and listing suggestions; recent searches; breadcrumbs; infinite scroll; a three-step create wizard that appends to local state; gallery thumbnails; phone reveal; store follow; FAQ accordion; an ads inquiry form; a responsive shell at an 860px breakpoint; hide-on-scroll header and bottom navigation; Escape-to-close; and both desktop and mobile mega-menus.

That is a real, thought-through user experience. **It is the most valuable thing in the folder** and should be treated as a frozen design specification alongside the docx.

2.5 Problems found in the mockup

ID	Severity	Finding
M1	Critical	No URLs. Routing is a <code>state.route</code> string. No deep links, no back button, no shareable listing links, nothing crawlable. For a classifieds site this is fatal and must be redesigned, not ported.
M2	Critical	No authentication anywhere. “Giriş” at line 63 calls <code>goProfile()</code> — it navigates straight into a logged-in profile. No login, register, OTP or session concept exists.
M3	High	Design tokens diverge from the document. The mockup uses <code>#315B48</code> rather than <code>#1F3A32</code> as the primary green, and <code>#FF5A1F</code> rather than <code>#D97732</code> as the accent — and applies that hot orange to links, hovers, active nav, focus rings and badges, contradicting §3’s “use sparingly”.
M4	High	Dead code. <code>homeTabs</code> , <code>tabItems</code> (§6 by-category section), <code>showGuide</code> (§6 outdoor guide), <code>footerCols</code> (§6 footer) and <code>wizPreview</code> (§14 preview step) are computed but referenced zero times in markup. Four documented sections are silently missing.
M5	High	There is no footer at all. §6 section 10 is unrendered.
M6	High	No hero. §8’s nature-image hero with the tagline is replaced by a compact search strip. The “outdoor premium” half of the brand thesis is not expressed on the homepage.
M7	Medium	Grid density wrong. <code>repeat(auto-fit, minmax(230px, 1fr))</code> with a 14px gap in a 1232px container yields 5 cards per row , not the specified 4, and there is no tablet-specific 3-per-row rule.
M8	Medium	Card omits category. §9 requires it; the card renders price, title and a merged “region, date, time” meta line only.

ID	Severity	Finding
M9	Legal	The privacy policy is headed " <i>İstifadəçi Razılaşmasına Əlavə №4</i> " — an appendix number to an agreement that does not exist here. Structure and phrasing read as adapted from another Azerbaijani platform's terms. Must be reviewed and rewritten by a lawyer before launch , not carried over.
M10	Legal	The ads page publishes concrete traffic claims — 38k monthly visitors, 7,400 active listings, 61% mobile, 2.4 min session — and a price list (900 / 520 / 1400 AZN). These are fabricated placeholders; publishing them as fact would misrepresent the platform to advertisers.
M11	Medium	Category photos hotlink <code>images.pexels.com</code> — a licensing and availability risk in production.
M12	Medium	Invented entities. Store rating 4.8, 214 reviews and <code>verified: true</code> are hardcoded. Reviews appear nowhere in the specification — the mockup invented a whole subsystem.
M13	Medium	Contact details <code>(012) 526-19-19</code> , <code>info@ovcuprim.az</code> and <code>reklam@ovcuprim.az</code> are placeholders appearing in roughly eight places. Need confirmation.
M14	Medium	8 top-level categories versus the document's 6 — the mockup promoted Optika and Bıçaq out of Ovçuluq. Needs a decision.
M15	Medium	<code>wizSteps</code> uses <code>Math.min(n, 7)</code> inside a 3-step wizard — leftover from the document's 8-step design. The wizard collapses §14's eight steps into three and drops the preview.
M16	Low	A custom crosshair cursor is forced on every element with <code>!important</code> — charming in a mockup, harmful to real usability and accessibility.
M17	Low	Desktop and mobile markup are largely duplicated (two headers, two category grids) rather than responsive, doubling maintenance.
M18	Low	<code>myRows</code> filters on <code>l.seller === 'Rəşad M. '</code> — a trailing-space bug that never matches; profile listings fall back to hardcoded IDs.
M19	Low	Accessibility: <code>aria-label</code> on icon buttons only; clickable <code>span / div</code> for nav and cards (no keyboard access); no landmarks; no focus management on the modal or drawer.
M20	Low	The gallery is fake — the counter is hardcoded to five images regardless of content.
M21	Low	<code>_ds/modernist-*</code> is an unrelated design system (red on white, Archivo, 0px radius, grayscale photography) that the canvas does not use. Leftover scaffold.

3 Current architecture assessment

There is no architecture to assess in the engineering sense. What exists is:

Presentation	One monolithic component, inline styles, no design-token layer, no component library, no reuse boundary except <code>ListingCard</code> .
State	A single 30-key object mixing routing, filters, wizard draft, UI toggles, favourites and per-store maps. No separation of server state from UI state.
Data	Module-level constants. No fetching, no loading states, no error states, no caching.
Routing	String comparison against <code>state.route</code> . No history integration.
Backend	None.
Persistence	None.
Auth & security	None.

VERDICT

The prototype is a validated design, not a foundation. Building the target system means starting a fresh repository and using `Ovcuprim.dc.html` as the visual and interaction reference.

4 Existing functionality assessment

Measured against the seventeen documented requirements, the prototype covers roughly **35% of the product surface** — and covers it only as design, with no data layer beneath it. The three most consequential absences are authentication (M2), URL-addressable routing (M1) and the entire server side. Everything the prototype does demonstrate, it demonstrates well; the gap analysis in section 6 states precisely where the line falls.

5 Reuse / Refactor / Rewrite / Remove analysis

Artifact	Disposition	Rationale
Prototype UX flows, IA, page composition	REUSE as specification	The real deliverable of the folder. Treat as the visual spec alongside the docx.
Prototype code	REWRITE	Proprietary runtime, inline styles, state-based routing. Nothing is portable to React/TypeScript.
<code>ListingCard.dc.html</code>	REWRITE (port faithfully)	Excellent card design; rebuild as a typed React component and add the missing category field.
CATS + 67 subcategories	REUSE → seed data	After reconciling with §5 (M14, D15). Becomes the Categories seed migration.

Artifact	Disposition	Rationale
REGIONS (12)	REFACTOR → seed data	Expand to the full region/district list; drop fabricated counts and compute them.
ATTRS	REFACTOR → seed data	Good starting point; becomes AttributeDefinitions. Needs types, units and options, not just placeholders.
HELP (22 FAQs)	REUSE (content)	High-quality Azerbaijani copy. Move to the database. Verify every rule it asserts is one you intend to enforce.
INFO — privacy & terms	REWRITE (legal review)	See M9. Do not ship.
INFO — other pages	REUSE with edits	Safe shopping, age/compliance, about — good drafts.
Ads pricing & statistics	REMOVE	See M10. Rebuild once real numbers exist.
L (24 listings)	REUSE → dev seed only	Useful for local development and tests. Never in production.
ICONS (8 SVG paths)	REUSE	Inline as a typed icon set.
Colour / type / radius tokens	REFACTOR	Reconcile M3 against §3 before writing any CSS.
support.js , image-slot.js	REMOVE	Canvas runtime, explicitly marked “do not edit”. No place in the product.
_ds/modernist-*	REMOVE	Unrelated design system (M21).
uploads/*.png	REVIEW → mostly REMOVE	40 sketch images. Keep as design reference outside the repo; none are production assets.
Canvas bookkeeping files	REMOVE	.image-slots.state.json , .thumbnail , scraps/ .
Router, auth, API, backend, DB, admin	CREATE	Nothing exists.

6 Requirement vs. implementation gap analysis

Already implemented — as design, against 24 mock rows

Category taxonomy and subcategories · listing grid with badges · client-side filtering across seven dimensions · three sort modes · search autocomplete with grouped suggestions · recent searches · breadcrumbs · favourites toggle and favourites page · detail page layout with gallery, specs, seller and similar listings · phone reveal · create-listing wizard · store list, store page and follow · profile with tabs and listing statuses · mobile bottom navigation · responsive shell · infinite scroll · FAQ and static-page systems · ads landing page.

Partially implemented — exists, needs completion

Requirement	Current state	What is missing
Homepage (§6)	5 of 10 sections	Hero, by-category tabs, region/map section, outdoor guide, footer, CTA band
Search (§11)	Substring match over 24 rows	Server-side execution, brand recognition, spelling tolerance, ranking, pagination
Filters (§10)	7 static filters	Delivery, date, brand, and all dynamic per-category attributes
Create wizard (§14)	3 steps	Real photo upload, preview step, validation, draft persistence, the documented 8-step flow
Detail page (§12)	Layout complete	Real gallery, working message button, working report, view counting
Store system (§13)	Aggregated from a seller string	Real Store entity, logo, address, description, rating source, verification
Profile (§19)	Static tabs	Authentication, real listings, edit/delete/archive/renew, settings
Listing card (§9)	Good	Category field; the 4/3/2 column rule
Map (§16)	Modal with region chips	An actual map
Moderation UI (§19)	Status badges only	Everything
SEO (§19)	Nothing	Everything

Missing entirely

Backend · database · REST API · authentication · authorization and roles · registration, login, logout · password or OTP handling · sessions and tokens · **image upload and storage** · thumbnails and optimisation · server-side pagination · admin panel · moderation queue · report handling · audit logging · listing lifecycle (expiry, renew, archive, quotas) · internal messaging · notifications · payments and promotions · blog/guide · footer · URL routing · server rendering and SEO · sitemap and robots · structured data · validation on both sides · error handling · logging · configuration and secrets · rate limiting · CORS · file-upload security · internationalisation · tests of any kind · CI/CD · deployment.

Incorrect — exists but does not satisfy the specification

M3 colour tokens · M6 no hero · M7 five columns instead of four · M8 no category on the card · M14 eight categories instead of six · M15 three wizard steps instead of eight, no preview · M2 “Giriş” is not a login · M12 invented ratings · M4

and M5 four documented sections silently dropped.

Technical debt to retire

M1 no routing · M16 forced cursor · M17 duplicated desktop/mobile markup · M18 seller-match bug · M19 accessibility · M20 fake gallery · M11 hotlinked images · M21 and the vendored runtime files · inline styles throughout · single-file monolith.

7 Target frontend architecture — React · Vite · TypeScript

Stack: React 19 · Vite 6 · TypeScript 5 with `strict` and `noUncheckedIndexedAccess` · React Router 7 · TanStack Query 5 for server state · react-hook-form with Zod · Tailwind CSS v4 driven by the documented tokens · Vitest and Testing Library · Playwright.

Deliberately excluded: Redux/MobX (TanStack Query plus URL state covers it), a component library (the design is bespoke), moment, lodash, axios.

```
frontend/
├── src/
│   ├── app/
│   │   ├── router.tsx           route tree, lazy boundaries
│   │   ├── providers.tsx       QueryClient, Auth, Toast, i18n
│   │   └── entry.client.tsx / entry.server.tsx
│   ├── layouts/
│   │   ├── PublicLayout.tsx    header + footer + mobile bottom nav
│   │   ├── AccountLayout.tsx  authenticated /kabinet shell
│   │   └── AdminLayout.tsx
│   ├── pages/                  one folder per route
│   ├── features/
│   │   ├── listings/          card, grid, gallery, detail blocks, create wizard
│   │   ├── search/            SearchBar, Autocomplete, RecentSearches
│   │   ├── filters/           FilterPanel, PriceRange, DynamicAttributeFilters,
│   │   │                       useFilterParams (URL ⇌ state)
│   │   ├── categories/ regions/ favorites/ stores/ auth/ moderation/ media/
│   │   └── components/ui/      Button, Input, Select, Chip, Badge, Card, Modal,
│   │                           Drawer, Skeleton, EmptyState, ErrorState, Pagination
│   ├── api/
│   │   ├── client.ts           fetch wrapper: auth header, refresh, ProblemDetails
│   │   ├── endpoints/*.ts      one module per resource
│   │   └── generated/types.ts   generated from the API's OpenAPI document
│   ├── hooks/ services/ types/ utils/ styles/ locales/
│   └── vite.config.ts tsconfig.json playwright.config.ts
```

Architectural rules

1. **The URL is the source of truth for search state.** Category, subcategory, region, price, condition, seller type, dynamic attributes, sort and page all live in the query string. This fixes M1 and makes results shareable and crawlable.
2. **Server state only through TanStack Query.** No listing data in component state.
3. **Every list view ships four states:** loading (a skeleton matching the real grid), empty, error with retry, and loaded.
4. **Zod schemas are the single validation contract**, mirrored server-side by FluentValidation.
5. **Mobile-first CSS** with one markup tree and responsive classes — retiring M17.

8 Target backend architecture — C# · ASP.NET Core · EF Core

Stack: .NET 9 · ASP.NET Core Web API · EF Core 9 with Npgsql · FluentValidation · Serilog · SixLabors.ImageSharp · Swashbuckle · xUnit with Testcontainers.

Four projects — separation without over-engineering. No MediatR, no CQRS split, and no repository layer over EF Core, because `DbContext` already is the unit of work.

```
backend/
├── src/
│   ├── Ovcuprim.Domain/           entities, enums, domain rules, no dependencies
│   ├── Ovcuprim.Application/      services (ListingService, SearchService, ...),
│   │                               DTOs, validators, interfaces (IFileStorage,
│   │                               ISmsSender, IImageProcessor), Result<T>
│   ├── Ovcuprim.Infrastructure/   AppDbContext, configurations, migrations,
│   │                               seed data, storage/SMS/image implementations
│   └── Ovcuprim.Api/              controllers, auth, middleware, DI, Program.cs
└── tests/ Domain.UnitTests · Application.UnitTests · Api.IntegrationTests
```

Errors	RFC 7807 <code>ProblemDetails</code> from global exception middleware. No stack traces in production.
Validation	FluentValidation on every request DTO, executed automatically.
Logging	Serilog structured JSON, request correlation IDs, PII scrubbed.
Configuration	<code>appsettings.json</code> plus environment variables. No secrets in source — user-secrets locally, environment or secret store in production.
Versioning	<code>/api/v1/...</code> from the first commit.

9 Database architecture

RECOMMENDATION — POSTGRESQL 16

Native full-text search with `pg_trgm` and `unaccent` covers requirement R9 (spelling tolerance) without a separate search cluster; `jsonb` with GIN indexes solves the dynamic-attribute problem (R8) cleanly; Npgsql gives first-class EF Core support; and it is the cheapest option to host well in this market. SQL Server is the alternative if existing licensing or Windows-hosting constraints apply.

Core entities

Entity	Fields, keys and constraints
Users	<code>Id</code> uuid PK · <code>PhoneNumber</code> varchar(20) unique · <code>Email</code> varchar(256) unique nullable · <code>PasswordHash</code> nullable · <code>FullName</code> varchar(100) · <code>Role</code> (User/Moderator/Admin) · <code>IsPhoneVerified</code> · <code>Status</code> (Active/Blocked/Deleted) · <code>CreatedAt</code> , <code>LastLoginAt</code> , soft-delete <code>DeletedAt</code>
RefreshTokens	<code>Id</code> · <code>UserId</code> FK cascade · <code>TokenHash</code> · <code>ExpiresAt</code> · <code>RevokedAt</code> · <code>ReplacedByTokenId</code> · <code>CreatedByIp</code> .Index (<code>UserId</code> , <code>ExpiresAt</code>)
OtpCodes	<code>PhoneNumber</code> · <code>CodeHash</code> · <code>Purpose</code> · <code>ExpiresAt</code> · <code>AttemptCount</code> · <code>ConsumedAt</code> .Index (<code>PhoneNumber</code> , <code>ExpiresAt</code>)
Categories	<code>Id</code> int PK · <code>ParentId</code> self FK restrict · <code>NameAz</code> / <code>NameRu</code> · <code>Slug</code> · <code>IconKey</code> · <code>SortOrder</code> · <code>IsActive</code> · <code>IsRestricted</code> (drives the D6 age/compliance gate) · <code>ListingCount</code> denormalised. Unique (<code>ParentId</code> , <code>Slug</code>) . Two levels only
AttributeDefinitions	<code>Id</code> · <code>CategoryId</code> FK · <code>Key</code> · <code>LabelAz</code> / <code>LabelRu</code> · <code>DataType</code> (Text/Number/Boolean/Enum) · <code>Unit</code> · <code>IsFilterable</code> · <code>IsRequired</code> · <code>SortOrder</code> .Unique (<code>CategoryId</code> , <code>Key</code>)
AttributeOptions	<code>Id</code> · <code>AttributeDefinitionId</code> FK cascade · <code>Value</code> · <code>LabelAz</code> / <code>LabelRu</code> · <code>SortOrder</code>
Regions	<code>Id</code> · <code>ParentId</code> self FK · <code>NameAz</code> / <code>NameRu</code> · <code>Slug</code> unique · <code>Latitude</code> / <code>Longitude</code> · <code>SortOrder</code>
Stores	<code>Id</code> · <code>OwnerUserId</code> FK unique (one store per user) · <code>Name</code> · <code>Slug</code> unique · <code>Description</code> · <code>LogoMediaId</code> FK nullable · <code>Address</code> · <code>Phone</code> · <code>IsVerified</code> · <code>Status</code> · <code>CreatedAt</code>
ListingMedia	<code>Id</code> · <code>ListingId</code> FK cascade · <code>StorageKey</code> · <code>Width</code> / <code>Height</code> / <code>SizeBytes</code> / <code>ContentType</code> · <code>SortOrder</code> · <code>IsPrimary</code> · <code>Variants</code> jsonb · <code>CreatedAt</code> .Unique (<code>ListingId</code> , <code>SortOrder</code>) ; check: max 10 rows per listing, exactly one primary
Favorites	Composite PK (<code>UserId</code> , <code>ListingId</code>) , both FK cascade · <code>CreatedAt</code> .Index (<code>UserId</code> , <code>CreatedAt</code> DESC)
StoreFollows	Composite PK (<code>UserId</code> , <code>StoreId</code>)
Reports	<code>Id</code> · <code>ListingId</code> FK · <code>ReporterUserId</code> FK nullable · <code>ReasonCode</code> · <code>Comment</code> · <code>Status</code> (Open/Reviewing/Resolved/Dismissed) · <code>ResolvedByUserId</code> · <code>ResolvedAt</code> .Index (<code>Status</code> , <code>CreatedAt</code>)

Entity	Fields, keys and constraints
ModerationActions	<code>Id</code> · <code>ListingId</code> FK · <code>ModeratorUserId</code> FK · <code>Action</code> · <code>Reason</code> · <code>CreatedAt</code> . Append-only
AuditLogs	<code>Id</code> · <code>ActorUserId</code> · <code>EntityType / EntityId</code> · <code>Action</code> · <code>PayloadJson</code> · <code>IpAddress</code> · <code>CreatedAt</code>
ListingQuotas	<code>Id</code> · <code>UserId</code> FK · <code>CategoryId</code> FK · <code>PeriodStart</code> date · <code>UsedCount</code> . Unique (<code>UserId</code> , <code>CategoryId</code> , <code>PeriodStart</code>) — implements the monthly free-listing rule
StaticPages	<code>Id</code> · <code>Slug</code> unique · <code>TitleAz / BodyAz / TitleRu / BodyRu</code> · <code>IsPublished</code> · <code>UpdatedAt</code>
FaqCategories / FaqItems	Mirrors the prototype's HELP structure
AdInquiries	<code>Id</code> · <code>Company</code> · <code>Contact</code> · <code>Note</code> · <code>Status</code> · <code>CreatedAt</code>

Listings — the hot table

`Id` uuid PK · `ShortId` bigint identity (for the SEO URL) · `Slug` varchar(120) · `UserId` FK restrict · `StoreId` FK nullable · `CategoryId` FK restrict · `RegionId` FK restrict · `Title` varchar(70) · `Description` varchar(3000) · `Price` numeric(12,2) · `Currency` char(3) default AZN · `Condition` · `HasDelivery` · `Brand` varchar(60) nullable · `SellerType` · `ContactPhone` · `ShowPhone` · `Status` (Draft / PendingModeration / Active / Rejected / Expired / Archived / Sold / Blocked) · `RejectionReason` · `Attributes` **jsonb** · `SearchVector` tsvector generated · `SearchKey` text (accent-folded, for trigram) · `PublishedAt` · `ExpiresAt` · `BumpedAt` · `ViewCount` · `FavoriteCount` · `CreatedAt` / `UpdatedAt` / `DeletedAt` .

Indexes: (`Status`, `BumpedAt` DESC) · (`CategoryId`, `Status`, `BumpedAt` DESC) · (`RegionId`, `Status`, `BumpedAt` DESC) · (`Status`, `Price`) · (`UserId`, `Status`) · **GIN** on `SearchVector` · **GIN** on `Attributes` `jsonb_path_ops` · **GIN trigram** on `SearchKey` · unique (`ShortId`) .

Relationships

User 1–0..1 Store · User 1–* Listing · Store 1–* Listing · Category 1–* Category (two levels) · Category 1–* Listing · Category 1–* AttributeDefinition · Region 1–* Listing · Listing 1–* ListingMedia · User *–* Listing via Favorites · User *–* Store via StoreFollows · Listing 1–* Report · Listing 1–* ModerationAction .

Business rules enforced at the data layer

One product per listing · title ≤ 70 characters, description ≤ 3,000 · between 1 and 10 images, ≤ 5 MB each, the first is primary · listings expire 30 days after publication · every new or edited listing enters PendingModeration · the phone is visible only when `ShowPhone` is set · restricted categories require an 18+ acknowledgement and extra moderation · deleting a user soft-deletes them and blocks their listings, while media is hard-deleted from storage when a listing is deleted.

Phase 2 tables: ListingPromotions (VIP/TOP/Frame with start and end dates) · Orders and Payments · Conversations and Messages · Notifications · SavedSearches · BlogPosts.

10 API architecture

Base path `/api/v1`. JSON, camelCase, offset pagination (`page` , `pageSize` ≤ 60) with the envelope `{ items, page, pageSize, total, totalPages }`, and errors as `ProblemDetails`.

Authentication

Method	Endpoint	Auth	Notes
POST	<code>/auth/register</code>	—	Phone + name (+ password if chosen); triggers OTP; rate-limited
POST	<code>/auth/verify-otp</code>	—	Consumes the code, issues tokens
POST	<code>/auth/login</code>	—	Phone + password, or phone + OTP
POST	<code>/auth/refresh</code>	cookie	Rotates the refresh token
POST	<code>/auth/logout</code>	✓	Revokes the token
POST	<code>/auth/forgot-password</code> · <code>/reset-password</code>	—	Rate-limited

Listings

Method	Endpoint	Auth	Notes
GET	<code>/listings</code>	—	The main query. Parameters: <code>q</code> , <code>categoryId</code> , <code>subCategoryId</code> , <code>regionId</code> , <code>minPrice</code> , <code>maxPrice</code> , <code>condition</code> , <code>sellerType</code> , <code>hasDelivery</code> , <code>brand</code> , <code>publishedWithin</code> , <code>attr.<key></code> , <code>sort</code> , <code>page</code> , <code>pageSize</code> . Server-side filter, sort and paginate. Returns Active only. Cached 60 s
GET	<code>/listings/{shortId}</code>	—	404 for non-active unless owner or moderator; increments view count asynchronously
GET	<code>/listings/{id}/similar</code>	—	Same category, excludes self, top 4
GET	<code>/listings/{id}/phone</code>	—	Returns the phone only when consent is given; rate-limited
POST	<code>/listings</code>	✓	Validates quota and required attributes, then enters moderation
PUT	<code>/listings/{id}</code>	✓ owner	Re-enters moderation
DELETE	<code>/listings/{id}</code>	✓ owner	Soft delete
POST	<code>/listings/{id}/archive</code> · <code>/restore</code> · <code>/renew</code> · <code>/mark-sold</code>	✓ owner	Lifecycle transitions
POST	<code>/listings/{id}/report</code>	optional	Creates a report; rate-limited by IP
GET	<code>/me/listings?status=</code>	✓	Backs the profile tabs

Media, taxonomy and search

Method	Endpoint	Auth	Notes
POST	/listings/{id}/media	✓ owner	Multipart, ≤10 files, ≤5 MB, magic-byte checked, re-encoded, variants generated
DELETE	/listings/{id}/media/{mediaId}	✓ owner	Removes every variant from storage
PUT	/listings/{id}/media/order	✓ owner	Reorder and set primary
GET	/categories · /categories/{slug} · /categories/{id}/attributes	—	Full tree, cached 1 h
GET	/regions · /regions/stats	—	Stats drive the map's per-region counts
GET	/search/suggest?q= · /search/popular	—	Grouped categories and listings, ≤200 ms budget, cached 5 min

Favourites, stores, profile, content

GET/POST/DELETE /favorites[//{listingId}] (auth) · GET /stores with search, filter and sort · GET /stores/{slug} · GET /stores/{slug}/listings · POST /stores (auth, enters verification) · PUT /stores/{id} · POST/DELETE /stores/{id}/follow · GET/PUT /me/profile · PUT /me/phone (re-verifies) · GET /pages/{slug} · GET /faq · POST /ad-inquiries (rate-limited, honeypot).

Administration

All under /admin/*, Moderator or Admin only: GET /admin/moderation/queue · POST /admin/listings/{id}/approve|reject|block · GET/PATCH /admin/reports · GET/PATCH /admin/users (block, unblock, role) · GET/POST/PUT/DELETE /admin/categories|regions|attributes|pages|faq · GET /admin/dashboard · GET /admin/audit-logs.

Every endpoint declares its authentication requirement, role requirement, validation rules and error shape in the OpenAPI document, which in turn generates the frontend types.

11 Page and route architecture

SEO-friendly Azerbaijani URLs. Every list route reads and writes its filter state to the query string.

Route	Page	Auth	API dependencies and notes
/	Home	—	categories, listings (featured + latest), region stats. Rebuild all 10 documented sections including hero and footer
/elanlar	All listings	—	/listings ; filters held in the URL
/elanlar/:categorySlug	Category	—	/listings + category attributes for dynamic filters
/elanlar/:categorySlug/:subSlug	Subcategory	—	As above
/axtaris?q=	Search results	—	/listings with relevance sort
/elan/:slug-:shortId	Listing detail	—	listing, similar, phone. Full meta tags and JSON-LD
/yeni-elan	Create wizard	✓	categories, attributes, media, listings. Draft kept in localStorage
/elan/:id/redakte	Edit listing	✓	Owner only; re-enters moderation
/secilmisler	Favourites	✓	/favorites
/magazalar	Store directory	—	/stores
/magaza/:slug	Store page	—	store + its listings
/kabinet	Profile overview	✓	/me/*
/kabinet/elanlarim	My listings	✓	Tabs: active, in moderation, archive
/kabinet/ayarlar	Settings	✓	Profile, phone change, account deletion
/giris · /qeydiyyat · /sifre-berpa	Authentication	—	/auth/* — all new
/melumat/:slug	Static pages	—	12 pages migrated from the prototype's INFO
/yardim	Help / FAQ	—	/faq
/reklam	Advertise	—	/ad-inquiries — real numbers only
/xerite	Map discovery	—	/regions/stats
/beledci · /beledci/:slug	Guide / blog	—	Pending decision D2
/admin/*	Admin panel	✓ role	/admin/* ; noindex
*	404	—	Suggests categories and search

Every page defines a loading skeleton, an empty state, an error state with retry, and mobile behaviour: bottom navigation visible, filters as a full-screen drawer, two-column grid.

12 Authentication & security strategy

- **Registration and login by phone number + SMS OTP** — the market norm and what the prototype's own help text promises. Password optional as a secondary factor. Requires an SMS gateway (Decision 4).
- **Passwords**, where used: ASP.NET Core Identity PBKDF2 or Argon2id. Never a hand-rolled hash.
- **Tokens**: a short-lived JWT access token (15 minutes) held in memory; a rotating refresh token (30 days, hashed at rest) in an **HttpOnly, Secure, SameSite=Strict** cookie. No token in `localStorage`.
- **Roles**: User, Moderator, Admin — policy-based authorization plus a resource-ownership check on every mutating listing and media endpoint.
- **Validation**: FluentValidation on the server, Zod on the client, one shared contract. The server never trusts the client.
- **Rate limiting**: OTP sends (3/hour/phone), login (10 per 15 minutes per IP), listing creation (per quota), report and phone-reveal endpoints, and search suggestions.
- **CORS**: an explicit origin allowlist with credentials enabled; no wildcards.
- **XSS**: React escapes by default; no `dangerouslySetInnerHTML` on user content; static-page HTML sanitised server-side; CSP with `default-src 'self'`.
- **SQL injection**: parameterised EF Core queries only; the one raw SQL query needed for full-text search uses parameters exclusively.
- **File upload**: extension, MIME and **magic-byte** validation; ImageSharp re-encode strips EXIF and any embedded payload; 5 MB and 10-file caps; randomised storage keys; served from a separate origin with no execute permission.
- **Secrets**: never in source or `appsettings.json`. User-secrets locally, environment variables or a secret store in production, `.env` gitignored.
- **Also**: HTTPS with HSTS, security headers, anti-automation honeypots on public forms, PII scrubbed from logs, and GDPR-style account deletion.

13 Search & filtering strategy

Server-side always — filtering, sorting and pagination all execute in PostgreSQL. The prototype's client-side approach breaks past a few hundred rows.

1. **Keyword**. A weighted `SearchVector` (title A, brand B, description C) using the `simple` configuration plus `unaccent`, ranked with `ts_rank_cd`.
2. **Spelling tolerance (R9/D7)**. Azerbaijani has no bundled PostgreSQL dictionary, so a normalisation function folds `ə→e, ı→i, ş→s, ç→c, ğ→g, ö→o, ü→u` into `SearchKey`, and `pg_trgm` similarity provides a fuzzy fallback when full-text returns too few rows. This handles both Azerbaijani diacritics and Russian-keyboard input.
3. **Brand recognition (R9/D8)**. A moderated Brand field with a seeded dictionary; an exact brand match short-circuits to a brand-filtered result set grouped by category — matching the document's "Shimano" example.
4. **Structured filters**. Indexed columns for category, subcategory, region, price, condition, seller type, delivery and date.
5. **Dynamic attributes (R8)**. `Attributes @> '{"length": "2.40"}'::jsonb` against the GIN index; the available filters and their options come from `AttributeDefinitions`, so adding a filter is a content change rather than a deployment.
6. **Sorting**. Newest, price ascending and descending, and relevance for keyword searches. Once VIP/TOP exists, promoted listings surface above organic results within each page.

- 7. **URL parameters.** Every filter is a query-string key and the page is fully reconstructible from the URL — this is the fix for M1.
- 8. **Result count and empty state.** Count via a windowed `COUNT(*) OVER()` ; the empty state offers “clear filters” plus nearest-region and category suggestions.
- 9. **Autocomplete.** Debounced 250 ms, previous request aborted, ≤ 200 ms server budget, results grouped as categories, listings and brands.
- 10. **Scale plan.** This design comfortably serves the low hundreds of thousands of listings. Only beyond that would Meilisearch or Elasticsearch be introduced — behind the existing `ISearchService` interface, so nothing else changes.

14 Media & image strategy

Upload	Multipart to <code>/listings/{id}/media</code> ; a client-side pre-check with an authoritative server-side re-check.
Limits	Max 10 images, 5 MB each — taken from the prototype's own help copy. JPEG, PNG, WebP and HEIC accepted.
Validation	Extension, declared MIME and magic bytes; dimension sanity bounds; animated or multi-frame payloads rejected.
Processing	ImageSharp re-encodes everything to WebP with a JPEG fallback, strips metadata, auto-rotates by EXIF, and emits four variants — thumb 160 ² , card 400×300, detail 1280w, og 1200×630.
Ordering	<code>SortOrder</code> with drag-reorder; the first image is primary and is used in cards, Open Graph tags and search results.
Deletion	Removing an image deletes every variant; deleting a listing deletes all its media; an orphan sweep runs nightly.
Storage	An <code>IFileStorage</code> abstraction with local disk for development and S3-compatible object storage in production (Cloudflare R2 recommended — no egress fees, CDN included). Swapping providers is a configuration change.
Delivery	CDN, immutable cache headers, content-hashed keys, lazy loading with explicit dimensions to prevent layout shift, and <code>srcset</code> per variant.
Security	Randomised keys (never user filenames), an origin separate from the API, no script execution, per-user upload rate limits. Retires M11 and M20.

15 Admin panel architecture

Served under `/admin` , marked `noindex` , and role-gated at both the router and the API.

Area	Admin	Moderator
Dashboard — listings by status, new users, pending queue, reports, 7/30-day trend	✓	read only
Moderation queue — approve, reject with reason, block; bulk actions; category and date filters	✓	✓
Listings — search all statuses, edit, force-archive, feature	✓	limited
Reports — triage, resolve, dismiss, act on the listing	✓	✓
Users — search, view, block/unblock, reset, change role	✓	view + block
Categories and attributes — CRUD, reorder, mark restricted	✓	✗
Regions — CRUD	✓	✗
Static pages and FAQ — CRUD	✓	edit
Media moderation — flagged images, bulk delete	✓	✓
Ad inquiries	✓	✗
Site settings — quotas, expiry days, image limits	✓	✗
Audit log — every moderation and admin action, immutable	✓	own actions

The moderation queue is the operational heart of the product. Per D6 and R2, listings in restricted categories route to a separate, stricter queue with a mandatory compliance checklist.

16 SEO strategy

THE CORE PROBLEM — D11 / M1

A purely client-rendered SPA will not rank for listing pages, and organic search is how classifieds platforms acquire both supply and demand. **Recommendation:** React Router 7 in framework (SSR) mode on a small Node server for public pages, with the authenticated dashboard and admin panel remaining client-only. This keeps the mandated React + Vite + TypeScript stack exactly as specified while making every public page server-rendered and crawlable.

Per page	Unique title (<code>{title}</code> – <code>{price}</code> AZN <code>Ovcuprim.az</code>), a generated meta description, a canonical URL (filters and pagination canonicalise to the clean category URL), Open Graph and Twitter cards using the primary image's og variant, and hreflang if Russian is added.
Structured data	Product and Offer on listing pages, BreadcrumbList everywhere, ItemList on category pages, Organization and WebSite/SearchAction on the homepage, FAQPage on help.

URLs	<code>/elan/karbon-spinning-tilov-240-m-12345</code> — transliterated slug plus short id; <code>/elanlar/baliqciliq/spinning</code> . Stable, lowercase, hyphenated, no IDs in category paths.
Crawl control	A dynamic sitemap index sharded by category (capped at 50k URLs per file), robots.txt disallowing admin, account and filtered permutations, noindex on empty result pages, pagination semantics via canonical, mandatory descriptive alt text, and 301s on slug changes.
Content	The guide section (R15) is the organic engine — treat it as a launch requirement rather than a nice-to-have if organic acquisition matters (see D2).

17 Performance strategy

Frontend: route-level code splitting; the create wizard, admin bundle and map lazy-loaded; lazy images with `srcset` and explicit dimensions; skeletons sized to real content (CLS < 0.1); debounced search; TanStack Query caching with a 5-minute stale time for taxonomy and 60 seconds for listings; listing detail prefetched on card hover; virtualisation only past roughly 200 items. **Budget: LCP < 2.5 s and TTI < 3.5 s on a mid-range Android over 4G, with initial JavaScript under 200 KB gzipped.**

Backend: server-side pagination with a hard page-size cap; projection to DTOs, never returning entities; `AsNoTracking` on all reads; the covering indexes listed in section 9; view counts incremented out of band; response compression; ETag and cache headers on taxonomy endpoints; an in-memory cache for categories, regions and popular searches (Redis only when scaling past one API instance); and `EXPLAIN ANALYZE` on the main listing query as a CI gate.

Infrastructure: CDN for all media and static assets, HTTP/2, Brotli, connection pooling, and a nightly job for expiry, count denormalisation and orphan-media sweeping.

18 Testing strategy

Frontend (Vitest + Testing Library): UI components including all four states; ListingCard rendering and favourite interaction; filter panel $\rightleftharpoons$ URL synchronisation; the create wizard's step logic and validation; Zod schemas; auth guards and redirects; API client error and token-refresh handling. Responsive behaviour is verified in Playwright at 375, 768 and 1440 pixels.

Backend (xUnit): unit tests for domain rules (expiry, quota, one-primary-image, status transitions), services with mocked storage and SMS, and validators; **integration tests with Testcontainers PostgreSQL** for every endpoint — auth flows, the full authorization matrix (anonymous, user, owner, moderator, admin), the search and filter query, pagination, media upload including malicious-file rejection, and moderation transitions.

End-to-end (Playwright) — the twelve critical flows: register with OTP and log in · create a listing with images · moderation approval takes it live · edit triggers re-moderation · delete and archive · keyword search · filter, sort and paginate · open a listing from search results · favourite and unfavourite · reveal the seller's phone · report a listing · admin moderates and blocks. These run against a seeded database in CI on every pull request.

Gates: at least 70% coverage on Application and Domain; all end-to-end tests green before deploy; Lighthouse ≥ 90 SEO and ≥ 85 performance on home, category and detail pages; an axe accessibility scan with zero criticals.

19 Development roadmap

Estimates assume **two developers** — one frontend, one backend. For a single developer, remove the parallelism and add roughly 60%.

Phase	Contents	Estimate
0 — Analysis & decisions	This document; answers to the approval checklist; resolution of D1–D16 and M9, M10, M13; final category tree and design tokens	now
1 — Foundation	Repository and solution scaffolding; Vite/React/TypeScript with strict config; the four .NET projects; PostgreSQL and the first migration; design-token layer and UI primitives; API client; CI pipeline; Docker Compose development environment	1–1.5 wk
2 — Authentication & users	Registration, OTP, login, refresh rotation, logout, password reset; roles and policies; profile endpoints; auth UI and guards; rate limiting. Closes M2	1.5–2 wk
3 — Taxonomy & content	Categories, subcategories, attribute definitions and regions — schema, seed, admin CRUD, public endpoints; static pages and FAQ migrated from the prototype with rewritten legal text	1–1.5 wk
4 — Listings & media	Listing CRUD, lifecycle and quotas; the image pipeline and storage; the create wizard with all eight documented steps including preview; the detail page; my-listings	2.5–3 wk
5 — Search & filtering	Full-text plus trigram plus jsonb attribute filters; autocomplete; sorting; pagination; the URL-driven filter panel; category and search pages. Closes M1	2–2.5 wk
6 — Discovery & user features	The full homepage with all ten sections including hero and footer ; favourites; store directory and store pages; follows; region and map discovery. Closes M4, M5, M6	2 wk
7 — Admin & moderation	Admin shell; the moderation queue including the restricted-goods flow; reports; user management; audit log; dashboard	2–2.5 wk
8 — SEO & performance	Server rendering for public routes; meta tags and JSON-LD; sitemap and robots; image and bundle optimisation; caching; index tuning; Lighthouse pass	1.5–2 wk
9 — Testing & QA	Complete the end-to-end suite; accessibility pass (closes M19); cross-browser and real-device testing; load-test the search query; security review	1.5–2 wk
10 — Production readiness	Deployment, TLS, backups with a restore drill, monitoring and alerting, error tracking, log aggregation, runbook, legal sign-off, content seeding	1–1.5 wk
Phase 2 — post-launch	Messaging · payments with VIP/TOP · verified sellers · notifications · blog and guide CMS · listing statistics · ad system · saved searches	separate

Total to production MVP: approximately 17–22 weeks with two developers. Phases 4/5 and 6/7 partially parallelise across the frontend and backend split.

20 Technical risks

#	Risk	Impact	Mitigation
R1	Legal exposure on regulated goods (D6) — hunting weapons, parts, certain knives	Site-level legal action	Restricted-category flag, mandatory 18+ gate, dedicated strict moderation queue, explicit prohibited-items policy, legal counsel before launch
R2	Copied legal text (M9)	Intellectual property / legal	Commission original terms and privacy policy
R3	Fabricated advertising metrics (M10)	Misrepresentation to advertisers	Remove until real analytics exist
R4	SPA versus SEO conflict (D11)	Little or no organic traffic — existential for classifieds	Server rendering from the start (Decision 1). Retrofitting it later is far more expensive
R5	SMS gateway cost and reliability	Blocks all registration	Choose a provider early; cache OTPs; rate-limit hard; keep email as a fallback
R6	The dynamic attribute schema is content work, not code (D10)	Filters ship empty	Author the 40+ subcategory attribute sets during Phases 3–4; assign a named owner
R7	Azerbaijani search quality (D7)	Users cannot find listings	Trigram plus folding from day one; log zero-result queries and tune
R8	Moderation throughput	Backlog; slow publication kills supply	Auto-approve trusted repeat sellers; keyword pre-screening; measure queue age
R9	Empty marketplace at launch	No demand without supply	Seed with real dealer and shop listings before opening publicly
R10	Image storage growth	Cost	WebP, variant caps, orphan sweep, and R2's zero egress fees
R11	Prototype scope is not production scope	Underestimation	The prototype covers roughly 35% of the system; this roadmap reflects the full 100%
R12	No migration risk — the project is greenfield	Low	The one real risk is losing the prototype's UX decisions; treat it as a frozen specification
R13	Internationalisation decided late (D12)	Schema rework	Decide before Phase 3; the proposed schema already carries Az and Ru columns
R14	Map data source undefined (D9)	Phase 6 slips	A static Azerbaijan SVG with region paths — no tile provider, no cost, no dependency
R15	Solo-developer bus factor	Delivery	Documented decisions, tests, CI, and no undocumented environment steps

21 Decisions requiring approval

Each stated as **Problem** → **Options** → **Recommendation** → **Reason**. Nothing here will be decided silently.

1. Rendering strategy — highest impact

Problem: The document requires SEO in the MVP (\$19) and a React SPA (\$18); these conflict.

Options: (a) SPA plus a prerender service; (b) React Router 7 server rendering; (c) meta injection for bot user-agents only.

Recommend (b).

Reason: Only real server rendering delivers correct crawling, fast first paint on mobile and working social previews, and it stays entirely within React + Vite + TypeScript. Option (a) is fragile; option (c) risks cloaking penalties.

2. Database

Problem: The document names none (D1).

Options: (a) PostgreSQL 16; (b) SQL Server; (c) MySQL.

Recommend (a).

Reason: Native full-text search and `pg_trgm` for Azerbaijani spelling tolerance, `jsonb` with GIN for dynamic attributes, no licence cost, and excellent Npgsql/EF Core support.

3. Category tree — six or eight top-level categories (M14, D15)

Options: (a) the document's six, with Optika and Bıçaq nested under Ovçuluq; (b) the mockup's eight.

Recommend (b), with the duplicates resolved.

Reason: Optics and knives cut across hunting, fishing and camping; nesting them under Hunting hides them from anglers.

4. Authentication method

Options: (a) email and password; (b) phone with SMS OTP — the prototype's promise and the market norm; (c) both.

Recommend (b), with email recovery.

Reason: Matches user expectation and gives free phone verification, which raises listing quality. It introduces a **recurring SMS cost requiring your budget approval and a provider choice**.

5. Languages (D12)

Options: (a) Azerbaijani only; (b) AZ + RU; (c) AZ + RU + EN.

Recommend: schema and UI built internationalisation-ready, ship Azerbaijani-only at launch, enable Russian in Phase 2.

Reason: Russian matters in Baku, but doubling all content and moderation at launch is a real cost. The proposed schema already carries both columns, so enabling Russian later is content work rather than a migration.

6. Blog / outdoor guide in the MVP (D2)

Options: (a) a full CMS in the MVP; (b) static guide pages in the MVP with a CMS in Phase 2; (c) Phase 2 entirely.

Recommend (b).

Reason: SEO content compounds slowly and should start early; a handful of hand-authored pages costs little and validates the format before an editor is built.

7. Shops in the MVP (D3)

Options: (a) full shops including verification in the MVP; (b) Store entity and storefront in the MVP, verification and paid packages in Phase 2; (c) Phase 2.

Recommend (b).

Reason: Shops are the supply side; the prototype already models them and the “Mağaza” badge drives buyer trust. Verification is process work that can follow.

8. Message button on the detail page (D4)

Options: (a) build messaging in the MVP, adding roughly three weeks; (b) remove the button and use phone contact only; (c) keep it as a “coming soon” stub.

Recommend (b).

Reason: A dead button erodes trust, and phone contact is the dominant channel in this market. Reintroduce it with real messaging in Phase 2.

9. Ratings and reviews (M12)

Problem: The mockup shows 4.8 stars and 214 reviews, but the document never mentions reviews.

Options: (a) build them; (b) remove them from the UI until Phase 2.

Recommend (b).

Reason: Reviews need an anti-abuse design — verified transactions, disputes — that neither document specifies, and fake stars are worse than none.

10. Design tokens (M3)

Problem: The mockup's greens and oranges differ from §3.

Options: (a) the document's values; (b) the mockup's values; (c) a hybrid.

Recommend (c): #1F3A32 as brand and header, the mockup's lighter #315B48 as the interactive green, and the document's #D97732 as the accent used sparingly per §3.

Reason: Preserves the documented brand while keeping the prototype's better contrast. The hot #FF5A1F applied everywhere directly contradicts the document's instruction.

11. Styling approach

Options: (a) Tailwind v4 with design tokens as CSS variables; (b) CSS Modules with tokens; (c) CSS-in-JS.

Recommend (a).

Reason: Zero runtime cost, enforces the token system, fastest for a bespoke design, and it is a build-time dependency — consistent with “avoid unnecessary frontend dependencies”. Option (b) is a fine alternative if plain CSS is preferred; (c) is ruled out for server-rendering performance.

12. Media storage

Options: (a) local disk on the VPS; (b) Cloudflare R2; (c) Azure Blob or AWS S3.

Recommend (b), behind an `IFileStorage` abstraction.

Reason: No egress fees, built-in CDN, S3-compatible — and the abstraction makes the choice reversible.

13. Deployment target

Options: (a) VPS with Docker Compose behind Cloudflare; (b) Azure App Service with Azure Database; (c) Kubernetes.

Recommend (a) for launch.

Reason: Dramatically lower cost at MVP traffic and simple to operate. Option (b) is the right upgrade for managed backups and scaling; (c) is over-engineering.

14. Restricted-goods policy (D6 / R1) — requires your answer, not mine

Problem: Which categories are permitted at all? Are firearms prohibited outright or permitted with documentation? What proof is required, and who bears liability?

My recommendation: prohibit firearms and their parts entirely at launch, and permit only unregulated accessories, optics and knives behind an 18+ gate.

Reason: This is a legal and business decision that must precede Phase 3, because it shapes both the category schema and the moderation workflow.

15. Placeholder content (M10, M13)

Confirm or replace: (012) 526-19-19, `info@ovcuprim.az`, `reklam@ovcuprim.az`, all advertising traffic statistics and prices, and the legal texts (M9).

16. Free-listing quota values

The prototype asserts a monthly per-category quota but never states the number. What is it, and is it per category or per account?

22 Approval checklist

Answering these unblocks Phase 1. Items marked **BLOCKING** stop the build; the rest have a stated default that will be used if you simply approve the recommendations.

#	Decision	Recommendation	Approved
1	Rendering strategy BLOCKING	React Router 7 server rendering for public pages	☐
2	Database BLOCKING	PostgreSQL 16	☐
3	Authentication BLOCKING	Phone + SMS OTP (provider and budget needed)	☐
4	Restricted-goods policy BLOCKING	Prohibit firearms and parts; 18+ gate on accessories	☐
5	Category tree	Eight top-level categories, duplicates resolved	☐
6	Languages	Azerbaijani at launch, schema Russian-ready	☐
7	Blog / guide in the MVP	Static guide pages now, CMS in Phase 2	☐
8	Shops in the MVP	Storefronts yes, verification in Phase 2	☐
9	Messaging	Remove the button; phone contact only in the MVP	☐
10	Ratings and reviews	Remove from the UI until Phase 2	☐
11	VIP / TOP and payments	Phase 2; the MVP shows featured slots only, unsold	☐
12	Design tokens	Hybrid palette per Decision 10	☐
13	Styling	Tailwind v4 with token variables	☐
14	Media storage	Cloudflare R2 behind <code>IFileStorage</code>	☐
15	Deployment	VPS with Docker Compose behind Cloudflare	☐
16	Listing lifecycle	30-day expiry, 10 images × 5 MB, 70/3000 character limits	☐
17	Free-listing quota	Your number is needed	☐
18	Legal texts (M9)	Rewrite with counsel — do not reuse	☐
19	Advertising statistics and prices (M10)	Remove until real data exists	☐
20	Contact details (M13)	Confirm or replace	☐
21	Map implementation	Static SVG of Azerbaijan by region	☐
22	Repository layout	Monorepo: <code>frontend/</code> , <code>backend/</code> , <code>docs/</code> ; git initialised in <code>D:\ovcu</code>	☐

#	Decision	Recommendation	Approved
23	Prototype disposal	Move the canvas folder to <code>docs/design-reference/</code> , excluded from the build	☐
24	Team size and timeline	Two developers, approximately 17–22 weeks to MVP	☐

STATUS

Analysis complete — awaiting approval. No files created in the project, no dependencies installed, no migrations run, and no implementation started.

Two notes alongside the approvals. First, the prototype's business rules — 30-day expiry, image limits, monthly quotas — currently exist only in the mockup's FAQ copy; they have been treated as authoritative here, so any that are wrong should be corrected before Phase 3. Second, decision 14 (restricted goods) genuinely should be settled with legal advice, since it shapes both the category schema and the moderation workflow.